

(16.4)

(a)

$$z = -1.836 + 3.021 \times 1.5 = 2.6955$$

$$p = \Lambda(2.6955) = \frac{1}{1 + e^{-2.6955}} \approx 0.9368 \quad \#$$

(b)

$$\text{threshold} = 0.5$$

由 (a) 知 $p = 0.9368$, 故 $\hat{y} = 1$

實際的 $y = 1$, 故 prediction 是正確的!

(c)

$$\begin{cases} x_1 = -1 \\ x_2 = -2 \end{cases} \quad z_i = -1 + 2x_i$$

$$i=1 \quad \begin{cases} x=1.5 \\ y=1 \end{cases} \rightarrow \begin{cases} z = -1 + 2 \times 1.5 = 2 \\ \Lambda(2) = \frac{1}{1 + e^{-2}} = 0.8808 \end{cases}$$

$$i=2 \quad \begin{cases} x=0.6 \\ y=1 \end{cases} \rightarrow \begin{cases} z = -1 + 2 \times 0.6 = 0.2 \\ \Lambda(0.2) = \frac{1}{1 + e^{-0.2}} = 0.5498 \end{cases}$$

$$i=3 \quad \begin{cases} x=0.7 \\ y=0 \end{cases} \rightarrow \begin{cases} z = -1 + 2 \times 0.7 = 0.4 \\ \Lambda(0.4) = \frac{1}{1 + e^{-0.4}} = 0.5938 \\ 1 - \Lambda(0.4) = 0.4062 \end{cases}$$

$$\ln L(x_1, x_2) = \sum_{i=1}^n (y_i \ln p(y_i=1|x_i) + (1-y_i) \ln(1-p(y_i=1|x_i)))$$

$$\rightarrow \ln L = \ln 0.8808 + \ln 0.5498 + \ln 0.4062 = -1.6378$$

最大似然估計值 $= -1.612$ 較大

因 MLE 使 log-likelihood 最大

① TRUE, 因 $\Lambda(z)^{y_i} (1-\Lambda(z))^{1-y_i}$ 為 probability $\rightarrow$ 相乘一定在 0 和 1 之間

② TRUE, 因 $\ln L = \sum_{i=1}^n (y_i \ln p(y_i=1|x_i) + (1-y_i) \ln(1-p(y_i=1|x_i))) = \sum_{i=1}^n \ln(p)$

又因 $p < 1$, 故 $\ln(p) < 0$, 加總後一定小於 0